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Packaged semiconductor device including heat slug

Abstract

This description relates generally to semiconductor devices and processes. A method for forming a packaged semiconductor package can include attaching a front side of a metal layer to a die pad of a leadframe that includes conductive terminals, so a periphery portion of the metal layer extends beyond a periphery pad surface of the die pad, and a portion of a half-etched cavity on the front side of the metal layer is located near the periphery pad surface of the die pad. The method further includes attaching a semiconductor device to the die pad and encapsulating the semiconductor device, the front side of the metal layer, a portion of a back side of the metal layer, and a portion of the conductive terminals to form a packaged semiconductor device.

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Background/Summary

TECHNICAL FIELD

(1) This description relates generally to semiconductor devices and processes, and more specifically, to methods of forming a semiconductor package.

BACKGROUND

(2) A semiconductor package is a metal, plastic, glass, or ceramic casing containing one or more discrete semiconductor devices or ICs and a leadframe. Leadframes are metal structures inside a semiconductor package that carry signals from a semiconductor device to an outside (e.g., external circuitry, such as to a printed circuit board (PCB)). A leadframe is composed of a central thermal die pad, where the semiconductor device is to be placed, and leads, which are metal structures for connecting the semiconductor device with the outside. The semiconductor device inside the semiconductor package can be attached to the central die pad by conductive (e.g. silver or solder paste) or non-conductive materials.

SUMMARY

(3) In an example, a method can include attaching a front side of a metal layer to a die pad of a leadframe that includes conductive terminals, so a periphery portion of the metal layer extends beyond a periphery pad surface of the die pad, and a portion of a half-etched cavity on the front side of the metal layer is located near the periphery pad surface of the die pad. The method further includes attaching a semiconductor device to the die pad and encapsulating the semiconductor device, the front side of the metal layer, a portion of a back side of the metal layer, and a portion of the conductive terminals to form a packaged semiconductor device.

(4) In a further example, a method can include etching grooves on a front side of a metal layer opposite of a back side of the metal layer, and attaching the front side of the metal layer to a die pad of a leadframe comprising conductive terminals, so a periphery portion of the metal layer extends beyond a periphery pad surface of the die pad, and the etched grooves are located near the periphery of the die pad. The method further includes attaching a semiconductor device to the die pad, and encapsulating the semiconductor device, the front side of the metal layer, a portion of the back side of the metal layer, and a portion of the conductive terminals to form a packaged semiconductor device.

(5) In another example, a semiconductor device can include a leadframe comprising a die pad. The die pad can include through holes extending through the die pad. The semiconductor device can further include a chip attached to the die pad, and a metal layer comprising grooves on a side of the metal layer, wherein a periphery portion of the metal layer extends beyond a periphery pad surface of the die pad, and the grooves are located near the periphery of the die pad. The semiconductor device includes a layer of a bonding material between the metal layer and the die pad.

(6) In another example, a semiconductor device can include a leadframe that can include a die pad. The die pad can include through holes extending through the die pad. The semiconductor device can include a chip attached to the die pad and a metal layer comprising protruding posts on a front side of the metal layer opposite of a back side of the metal layer, in which the protruding posts extend into respective through holes of the die pad, and a periphery portion of the metal layer extends beyond a periphery pad surface of the die pad.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an example of a cross-sectional view of a semiconductor device.

(2) FIG. 2 is an example of a cross-sectional view of another semiconductor device.

(3) FIG. 3 is an example of a leadframe assembly with a heat slug for a semiconductor device.

(4) FIG. 4 is an example of a molded package with a heat slug.

(5) FIG. 5 is an example of a top view of a leadframe assembly for a semiconductor device.

- (6) FIG. 6 is an example of a top view of another leadframe assembly for a semiconductor device.
- (7) FIG. 7 is an example of a top view of a further leadframe assembly for a semiconductor device.
- (8) FIG. 8 is an example of a top view of a yet another leadframe assembly for a semiconductor device.
- (9) FIG. 9 is an example of a partial view of a semiconductor device.
- (10) FIG. 10 is an example of a first stage of a process flow for fabricating a metal layer of a semiconductor device.
- (11) FIG. 11 is an example of a second stage of a process flow for fabricating a leadframe assembly of the semiconductor device.
- (12) FIG. 12 is an example of a third stage of a process flow for fabricating the semiconductor device that includes attaching the metal layer to the leadframe assembly.
- (13) FIG. 13 is an example of a fourth stage of a process flow for fabricating the semiconductor device that includes applying an adhesive material for attaching a chip.
- (14) FIG. 14 is an example of a fifth stage of a process flow for fabricating the semiconductor device that includes attaching the chip to the leadframe assembly.
- (15) FIG. 15 is an example of a sixth stage of a process flow for fabricating the semiconductor device that includes attaching at least one wire to a chip bonding pad of the chip and a respective conductive terminal of conductive terminals of the leadframe assembly.
- (16) FIG. 16 is an example of a seventh stage of a process flow for fabricating the semiconductor device that including applying a molding compound.
- (17) FIG. 17 is an example of a first stage of another process flow for fabricating a metal layer of a semiconductor device.
- (18) FIG. 18 is an example of a second stage of another process flow for fabricating a leadframe assembly of the semiconductor device.
- (19) FIG. 19 is an example of a third stage of another process flow for fabricating the semiconductor device that includes applying an adhesive material for attaching the metal layer to the leadframe assembly.
- (20) FIG. 20 is an example of a fourth stage of another process flow for fabricating the semiconductor device that includes attaching the metal layer to the leadframe assembly.
- (21) FIG. 21 is an example of a fifth stage of another process flow for fabricating the semiconductor device that includes applying an adhesive material for attaching a chip.
- (22) FIG. 22 is an example of a sixth stage of another process flow for fabricating the semiconductor device that includes attaching the chip to the leadframe assembly.
- (23) FIG. 23 is an example of a sixth stage of a process flow for fabricating the semiconductor device that includes attaching at least one wire to a chip bonding pad of the chip and a respective conductive terminal of conductive terminals of the leadframe assembly.
- (24) FIG. 24 is an example of a seventh stage of another process flow for fabricating the semiconductor device that includes applying a molding compound.

DETAILED DESCRIPTION

(25) This description relates generally to semiconductor devices and processes. High power ICs in quad flat non-leaded (QFN) packages are becoming smaller and heat generated by one or more chips packaged therein needs to be adequately dissipated to avoid impact on chip performance. One approach for dissipating the heat generated by the one or more chips within the QFN package is to use thicker leadframes in contrast to leadframes that are used in standard QFN packages. However, the use of thicker leadframes increases a size of the QFN package, which increases an overall unit cost of the QFN package. Alternatively, a different package can be used, such as a transistor outline (TO) package, however, these packages do not have sufficient pins for power circuitry (e.g., power stage circuitry) in contrast to a QFN package.

(26) In some examples, a semiconductor device can be provided with a heat slug. During fabrication of the semiconductor device, the heat slug can be attached to a patterned formed

leadframe, which can be packaged with a chip to provide a packaged semiconductor device. For example, the heat slug can be attached to a front side of a formed central thermal die pad of the patterned formed leadframe opposite of a back side to which at least one chip is attached to increase a heat dissipation area for the at least one chip attached to the formed central thermal die pad. In some examples, the patterned formed leadframe can have a single gauge leadframe (e.g., in which the formed central thermal die pad and conductive leads of the patterned formed leadframe have about a same thickness). The term “heat slug” as used herein can refer to a metal layer that has a greater surface area than a formed central thermal die pad of a patterned formed leadframe to which the heat slug is to be attached.

(27) By attaching the heat slug to the patterned formed leadframe a heat dissipation area for the chip can be increased, thereby improving a thermal performance of the semiconductor device. The heat slug can be attached to the formed central thermal die pad to provide an increased area for spreading heat produced by one or more chips therein. The examples herein illustrate heat slugs attached to patterned formed leadframes and packaged in QFN packages. In other examples, a different package can be used, for example, other types of no-lead packages, such as Small Outline No-Lead (SON) can be used for packaging leadframes.

(28) FIG. 1 is an example of a cross-sectional view of a semiconductor device **100**. The semiconductor device **100** can be a packaged QFN-type semiconductor device. The semiconductor device **100** can include conductive terminals **102** and a formed central thermal die pad **104**. The conductive terminals **102** and the formed central thermal die pad **104** can form a leadframe **106** for a chip **108** of the semiconductor device **100**. As an example, the chip **108** can be a driver IC, such as a Gallium Nitride (GaN) gate driver. In other examples, the chip **108** may be a different type of chip. The formed central thermal die pad **104** can include through holes **110**. The through holes **110** can extend through the formed central thermal die pad **104**.

(29) In some examples, the semiconductor device **100** includes a heat slug **112**. The heat slug **112** can have a greater surface area and in some examples a greater thickness than the formed central thermal die pad **104**, as shown in FIG. 1. In some examples, the heat slug **112** has a similar thickness as the leadframe **106**. In examples, wherein the heat slug **112** and the leadframe **106** have a similar thickness, the semiconductor device **100** can be QFN packaged while retaining a similar thickness package thickness as a standard QFN package. As an example, the standard QFN package can have a thickness of about 1 millimeter (mm) to about 5 mm. The heat slug **112** can include protruding posts **114** on a first side **116** of the heat slug **112** opposite of a second side **118** of the heat slug **112**. The protruding posts **114** can extend into respective through holes **110** of the formed central thermal die pad **104**. A periphery portion **120** of the heat slug **112** can extend beyond a periphery pad surface of the formed central thermal die pad **104**. The heat slug **112** further includes a half etched-cavity **122** on the first side **116** of the heat slug **112** and a portion of the half-etched cavity **122** can be located near a periphery of the formed central thermal die pad **104**. In some examples, a gap **124** can separate at least some of the conductive terminals **102** from the formed central thermal die pad **104**.

(30) In some examples, a riveting process can be used to secure the heat slug **112** to the formed central thermal die pad **104**. For example, during the riveting process, each of the protruding posts **114** can be inserted through a respective through hole of the through holes **110**, such that a respective tail end **126** of each protruding post extends through the respective through hole. In other examples, a laser welding process can be used to secure the heat slug **112** to the formed central thermal die pad **104**. For example, during the laser welding process, each of the protruding posts **114** can be inserted through the respective through hole of the through holes **110**, such that the respective tail end **126** of each protruding post extends through the respective through hole. In either of the riveting or laser welding processes, the tail ends **126** of the respective protruding posts **114** can be deformed to secure the heat slug **112** to the formed central thermal die pad **104**.

(31) In some examples, the semiconductor device **100** includes at least one wire **128**. As an

example, the at least one wire **128** can be a copper wire. The at least one wire **128** can span the gap **124** between a chip bonding pad **130** of the chip **108** and a respective conductive terminal of the conductive terminals **102**. For clarity and brevity purposes, the example of FIG. **1** shows a single wire spanning the gap **124**, however, it is be understood that any number of wires may span the gap **124** between a respective chip bonding pad of the chip **108** and the respective conductive terminal of the conductive terminals **102**. In some examples, the semiconductor device **100** includes a packaging material **132**. The packaging material **132** can be a molding compound and can encapsulate the chip **108**, the first side **116** of the heat slug **112**, a portion of the conductive terminals **102**, and a portion of the second side **118** of the heat slug **112** to provide a packaged semiconductor device. In some examples, the chip **108** can be attached via a bonding material **134** to a surface of the formed central thermal die pad **104** opposite to a surface to which the heat slug **112** is attached.

(32) Because the heat slug **112** has a greater surface area than the formed central thermal die pad **104**, attaching the heat slug **112** to the formed central thermal die pad **104** increases a heat spreading area for spreading heat produced by the chip **108**. The heat produced by the chip **108** can be provided conductively to the formed central thermal die pad **104**, which can provide the heat to the heat slug **112** for dissipation into an environment or atmosphere. Accordingly, the packaged QFN-type semiconductor device **100** can have a lower thermal resistance in contrast to existing packaged QFN-type semiconductor devices.

(33) FIG. **2** is an example of a cross-sectional view of another semiconductor device **200**. The semiconductor device **200** can be a packaged QFN-type semiconductor device. The semiconductor device **200** can include conductive terminals **202** and a formed central thermal die pad **204**. The conductive terminals **202** and the formed central thermal die pad **204** can form a leadframe **206** for a chip **208** of the semiconductor device **200**. As an example, the chip **208** can be a driver IC, such as a GaN gate driver. In other examples, the chip **208** may be a different type of chip.

(34) In some examples, the semiconductor device **200** includes a heat slug **210**. The heat slug **210** can have a greater surface area and in some examples a greater thickness than the formed central thermal die pad **204**, as shown in FIG. **2**. In some examples, the heat slug **210** has a similar thickness as the leadframe **206**. In examples wherein the heat slug **210** and the leadframe **206** have a similar thickness, the semiconductor device **200** can be QFN packaged while retaining a similar thickness package thickness as a standard QFN package. In some examples, the heat slug **210** can be attached via a bonding material **212** to a first surface of the formed central thermal die pad **204** opposite a second surface. The heat slug **210** can include openings **214** on a first side **216** opposite of a second side **218** of the heat slug **210**. In some examples, the openings **214** are grooves, such as half-etched grooves.

(35) The openings **214** can be used during fabrication or assembly of the semiconductor device **200** for guidance (e.g., placement) of the chip **208** on the first surface of the formed central thermal die pad **204**. The openings **214** can be located near a periphery portion **220** of the heat slug **210**. As shown in FIG. **2**, the periphery portion **220** of the heat slug **210** can extend beyond a periphery pad surface of the formed central thermal die pad **204**. In some examples, a gap **222** can separate at least some of the conductive terminals **202** from the formed central thermal die pad **204**. The openings **214** can be at least partially located within the gap **222**, and thus at about the periphery portion **220** of the heat slug **210**.

(36) In some examples, the semiconductor device **200** includes at least one wire **224**. As an example, the at least one wire **224** can be a copper wire. The at least one wire **224** can span the gap **222** between a chip bonding pad **226** of the chip **208** and a respective conductive terminal of the conductive terminals **202**. For clarity and brevity purposes, the example of FIG. **2** shows a single wire spanning the gap **222**, however, it is be understood that any number of wires may span the gap **222** between a respective chip bonding pad of the chip **208** and the respective conductive terminal of the conductive terminals **202**. In some examples, the semiconductor device **200** includes a

packaging material **228**. The packaging material **228** can be a molding compound and can encapsulate the chip **208**, the first side **216** of the heat slug **210**, a portion of the conductive terminals **202**, and a portion of the second side **218** of the heat slug **210** to provide a packaged semiconductor device. In some examples, the chip **208** can be attached via a bonding material **230** to the second surface of the formed central thermal die pad **204** opposite the first surface to which the heat slug **210** is attached.

(37) Because the heat slug **210** has a greater surface area than the formed central thermal die pad **204**, attaching the heat slug **210** to the formed central thermal die pad **204** increases a heat spreading area for spreading heat produced by the chip **208**. The heat produced by the chip **208** can be provided conductively to the formed central thermal die pad **204**, which can provide the heat to the heat slug **210** for dissipation into an environment or atmosphere. Accordingly, the packaged QFN-type semiconductor device **200** can have a lower thermal resistance in contrast to existing packaged QFN-type semiconductor devices.

(38) FIG. **3** is an example of leadframe assembly **300** of a semiconductor device, such as the semiconductor device **100**, as shown in FIG. **1**, or the semiconductor device **200**, as shown in FIG. **2**. Thus, reference can be made to the example of FIGS. **1-2** in the example of FIG. **3**. The leadframe assembly **300** can include conductive terminals **302**, a formed central thermal die pad **304**. As shown in FIG. **3**, a heat slug **306** (e.g., the heat slug **112**, as shown in FIG. **1**, or the heat slug **210**, as shown in FIG. **2**) can be coupled to the leadframe assembly **300**. In the example of FIG. **3**, the heat slug **306** is attached to a surface of the formed central thermal die pad **304** opposite a surface to which a chip (e.g., the chip **108**, as shown in FIG. **1**, or the chip **208**, as shown in FIG. **1**) can be attached.

(39) FIG. **4** is an example of a molded package **400**. The molded package **400** can include semiconductor device **100**, as shown in FIG. **1**, or the semiconductor device **200**, as shown in FIG. **2**. In some examples, the molded package **400** includes the leadframe assembly **300**, as shown in FIG. **3**. Thus, reference can be made to the example of FIGS. **1-3** in the example of FIG. **4**. As shown in FIG. **4**, the molded package **400** includes a packaging material **402** (e.g., the packaging material **132**, as shown in FIG. **1**, or the packaging material **228**, as shown in FIG. **2**), conductive leads **404** (e.g., the conductive terminals **102**, as shown in FIG. **1**, or the conductive terminals **202**, as shown in FIG. **2**), and a heat slug **406** (e.g., the heat slug **112**, as shown in FIG. **1**, or the heat slug **210**, as shown in FIG. **2**). The packaging material **402** can encapsulate a portion of the conductive leads **404** of the molded package **400**, and a portion of a second side **408** of the heat slug **406** of the molded package **400**. The second side **408** of the heat slug **406** can correspond to the second side **118**, as shown in FIG. **1**, or the second side **218**, as shown in FIG. **2**.

(40) FIG. **5** is an example of a top view of a leadframe assembly **500** for a semiconductor device, such as the semiconductor device **100**, as shown in FIG. **1**, or the packaged semiconductor device **400**, as shown in FIG. **4**. In some examples, the leadframe assembly **500** can be the leadframe assembly **300**, as shown in FIG. **3**. Thus, reference can be made to the example of FIGS. **1** and **3-4** in the example of FIG. **5**. The leadframe assembly **500** can include conductive terminals **502** extending around a perimeter of the leadframe assembly **500**, a formed central thermal die pad **504**, and a heat slug **506** (e.g., the heat slug **112**, as shown in FIG. **1**, or the heat slug **210**, as shown in FIG. **2**).

(41) In the example of FIG. **5**, the heat slug **506** is attached to a surface of the formed central thermal die pad **504** opposite a surface to which a chip (e.g., the chip **108**, as shown in FIG. **1**, or the chip **208**, as shown in FIG. **1**) can be attached. For example, protruding posts (e.g., the protruding posts **114**, as shown in FIG. **1**) of the heat slug **506** can be positioned through a respective through hole of through holes **508**, such that a distal end of each protruding post extends through the respective through hole. The heat slug **506** can be attached to the formed central thermal die pad **504**, such that a periphery portion **510** of the heat slug **506** extends beyond a periphery pad surface **512** of the formed central thermal die pad **504**. In some examples, the

periphery portion **510** as shown in FIG. 5 can correspond to the periphery portion **120**, as shown in FIG. 1.

(42) FIG. 6 is an example of a top view of another leadframe assembly **600** for a semiconductor device, such as the semiconductor device **100**, as shown in FIG. 1, or the packaged semiconductor device **400**, as shown in FIG. 4. In some examples, the leadframe assembly **600** can be the leadframe assembly **300**, as shown in FIG. 3. Thus, reference can be made to the example of FIGS. 1 and 3-4 in the example of FIG. 5. The leadframe assembly **600** can include conductive terminals **602** extending around a perimeter of the leadframe assembly **600**, a formed central thermal die pad **604**, and a heat slug **606** (e.g., the heat slug **112**, as shown in FIG. 1, or the heat slug **210**, as shown in FIG. 2).

(43) In the example of FIG. 6, the heat slug **606** is attached to a surface of the formed central thermal die pad **604** opposite a surface to which a chip (e.g., the chip **108**, as shown in FIG. 1, or the chip **208**, as shown in FIG. 1) can be attached. For example, protruding posts (e.g., the protruding posts **114**, as shown in FIG. 24) of the heat slug **606** can be positioned through a respective through hole of through holes **608**, such that a distal end of each protruding post extends through the respective through hole. As shown in FIG. 6, a periphery portion **610** of the heat slug **606** can extend beyond a periphery pad surface **612** of the formed central thermal die pad **604**. In some examples, the periphery portion **610** as shown in FIG. 6 can correspond to the periphery portion **120**, as shown in FIG. 1. The heat slug **606** can be formed with protruding posts that once positioned through the respective through hole of the through holes **608** offsets the heat slug **606** relative to the formed central thermal die pad **604**. The heat slug **606** can be offset relative to the formed central thermal die pad **604** to provide sufficient creepage clearance between the heat slug **606** and at least some of the conductive terminals **602**.

(44) FIG. 7 is an example of a top view of a leadframe assembly **700** for a semiconductor device, such as the semiconductor device **200**, as shown in FIG. 2, or the packaged semiconductor device **400**, as shown in FIG. 4. In some examples, the leadframe assembly **500** can be leadframe assembly **300**, as shown in FIG. 3. Thus, reference can be made to the example of FIGS. 2-4 in the example of FIG. 7. The leadframe assembly **700** can include conductive terminals **702** extending around a perimeter of the leadframe assembly **700**, a formed central thermal die pad **704**, and a heat slug **706** (e.g., the heat slug **112**, as shown in FIG. 1, or the heat slug **210**, as shown in FIG. 2).

(45) In the example of FIG. 7, the heat slug **706** is attached to a surface of the formed central thermal die pad **704** opposite a surface to which a chip (e.g., the chip **108**, as shown in FIG. 1, or the chip **208**, as shown in FIG. 1) can be attached. For example, the heat slug **706** can be attached via a bonding material (e.g., the bonding material **212**, as shown in FIG. 2) to a first surface of the formed central thermal die pad **704** opposite a second surface to which the chip can be attached. The heat slug **706** can be attached to the formed central thermal die pad **704**, such that a periphery portion **708** of the heat slug **706** extends beyond a periphery pad surface **710** of the formed central thermal die pad **704**. In some examples, the periphery portion **708** of FIG. 7 can correspond to the periphery portion **220**, as shown in FIG. 2.

(46) FIG. 8 is an example of a top view of a leadframe assembly **800** for a semiconductor device, such as the semiconductor device **200**, as shown in FIG. 2, or the packaged semiconductor device **400**, as shown in FIG. 4. In some examples, the leadframe assembly **800** can be the leadframe assembly **300**, as shown in FIG. 3. Thus, reference can be made to the example of FIGS. 2-4 in the example of FIG. 8. The leadframe assembly **800** can include conductive terminals **802** extending around a perimeter of the leadframe assembly **800**, a central thermal die pad **804**, and a heat slug **806** (e.g., the heat slug **112**, as shown in FIG. 1, or the heat slug **210**, as shown in FIG. 2).

(47) In the example of FIG. 8, the heat slug **806** is attached to a surface of the formed central thermal die pad **804** opposite a surface to which a chip (e.g., the chip **108**, as shown in FIG. 1, or the chip **208**, as shown in FIG. 1) can be attached. For example, the heat slug **806** can be attached via a bonding material (e.g., the bonding material **212**, as shown in FIG. 2) to a first surface of the

formed central thermal die pad **804** opposite a second surface to which the chip can be attached. The heat slug **806** can be attached to the formed central thermal die pad **804**, such that a periphery portion **808** of the heat slug **806** extends beyond a periphery pad surface **810** of the formed central thermal die pad **704**. The heat slug **606** can be attached to the formed central thermal die pad **804**, such that the heat slug **806** is offset relative to the formed central thermal die pad **804**. The heat slug **806** can be offset relative to the formed central thermal die pad **804** to provide sufficient creepage clearance between the heat slug **806** and at least some of the conductive terminals **802**. (48) FIG. **9** is an example of a semiconductor device **900** prior to being encapsulated with a packaging material to provide a packaged semiconductor device, such as the semiconductor device **100**, as shown in FIG. **1**, or the packaged semiconductor device **400**, as shown in FIG. **4**. The semiconductor device **900** can include a leadframe assembly **902**. The leadframe assembly **902** can be similar to the leadframe assembly **300**, as shown in FIG. **3**, or the leadframe assembly **500**, as shown in FIG. **5**. Thus, reference can be made to the example of FIGS. **1** and **4-5** in the example of FIG. **9**. The leadframe assembly **902** can include conductive terminals **904**, a formed central thermal die pad **906**, and a heat slug **908**.

(49) The heat slug **908** is attached to a surface of the formed central thermal die pad **906** opposite a surface **910** to which a chip **912** (e.g., the chip **108**, as shown in FIG. **1**) is attached. For example, protruding posts **914** of the heat slug **908** can be positioned through a respective through hole of through holes **916** of the formed central thermal die pad **906**, such that a distal end of each protruding post extends through the respective through hole. The heat slug **908** can be attached to the formed central thermal die pad **906**, such that a periphery portion **918** of the heat slug **908** extends beyond a periphery portion **920** of the formed central thermal die pad **906**. The chip **912** can be attached to the formed central thermal die pad **906** prior to encapsulation of the leadframe assembly **902** to provide a packaged semiconductor device. By way of example, as shown in FIG. **9**, the chip **912** can include a gate driver **922** for driving a GaN field effect transistor (FET) **924** of the chip **912**. Because in the example of FIG. **9** the heat slug **908** has a greater surface area than the formed central thermal die pad **904**, the heat slug **908** provides an increased heat spreading area for heat produced by the chip **912**. Heat produced by the chip **912** can be transferred conductively to the formed central thermal die pad **906**, which can transfer the heat to the heat slug **908** for dissipation into an environment.

(50) FIGS. **10-16** show stages of a method for fabricating a semiconductor device, such as the semiconductor device **100**, as shown in FIG. **1**. Thus, reference can be made to the example of FIG. **1** in the example of FIGS. **10-16**. As shown in FIG. **10**, at **1000**, in a first stage, a half-etched cavity **1002** can be etched on a first side **1004** opposite a second side **1006** of a first metal layer **1008**. In some examples, the first metal layer **1008** can be referred to as a heat slug (e.g., the heat slug **112**, as shown in FIG. **1**). By way of example, one or more masks can be deposited on the first metal layer **1008** and etched to provide the half-etched cavity **1002** and protruding posts **1012** therein. The one or more masks can be stripped away, so as to result in a structure as shown in FIG. **10**. The first metal layer **1008** may be a conductive metal, such as Copper. In some examples, the protruding posts **1012** can be formed on the first side **1004** of the first metal layer **1008**. In some examples, a first subset of the protruding posts **1012** can have a greater width than a second subset of the protruding posts **1012**, as shown in FIG. **10**. In other examples, the protruding posts **1012** can have a similar width. The protruding posts **1012** can be located within the half-etched cavity **1002**.

(51) As shown in FIG. **11**, at **1100**, in a second stage, a second metal layer **1014** can be patterned to provide a leadframe **1016** that includes a formed central thermal die pad **1018** and conductive terminals **1020**. For example, at **1100**, in the second stage, the second metal layer **1014** can be etched to include through holes **1022** in the formed central thermal die pad **1018**. In some examples, a first subset of the through holes **1022** can have a greater width than a second subset of through holes **1022**, as shown in FIG. **11**. The first subset of the through holes **1022** can have a

width that is greater than the width of the first subset of the protruding posts **1012** and the second subset of the through holes **1022** can have a width that is greater than the width of the second subset of protruding posts **1012**, as shown in FIGS. **10-11**. In other examples, the first and second subset of through holes **1022** can have a similar width as the first and second subset of the protruding posts **1012**. The second metal layer **1014** may be a conductive metal, such as Copper or Copper alloy. In some examples, the first metal layer **1008** can have a greater thickness than the second metal layer **1014**. In other examples, the first and second metal layers **1008** and **1014** can have a similar thickness. For example, the leadframe **1016** can be produced, at **1100**, in the second stage, by a chemical etching technique or a stamping technique. In other examples, the through holes **1022** can be produced after fabrication of the leadframe **1016**. As an example, a mask can be deposited on the formed central thermal die pad **1018** and etched to expose the through holes **1022**. The mask thereafter can be stripped away, so as to result in a structure as shown in FIG. **11**.

(52) As shown in FIG. **12**, at **1200**, in a third stage, the first metal layer **1008** can be attached to a first side **1024** opposite a second side **1026** of the formed central thermal die pad **1018**, so a periphery portion **1028** of the first metal layer **1008** extends beyond a periphery portion **1030** of the formed central thermal die pad **1018**, and at least some of the half-etched cavity **1002** is located near the periphery portion **1030** the formed central thermal die pad **1018**. At **1200**, in the third stage, the first metal layer **1008** can be attached to the first side **1024** of the formed central thermal die pad **1018**, such that the protruding posts **1012** extend into respective through holes **1022** of the formed central thermal die pad **1018**. In some examples, at **1200**, in the third stage, a riveting process can be used to secure the first metal layer **1008** to the formed central thermal die pad **1018** after positioning each of the protruding posts **1012** through a respective through hole of the through holes **1022**. For example, during the riveting process, each of the protruding posts **1012** can be inserted through a respective through hole of the through holes **1022**, such that a distal end of each protruding post extends through the respective through hole. In other examples, at **1200**, in the third stage, a laser welding process can be used to secure the first metal layer **1008** to the formed central thermal die pad **1018**. For example, during the laser welding process, each of the protruding posts **1012** can be inserted through a respective through hole of the through holes **1022**, such that a distal end of each protruding post extends through the respective through hole. During the riving or laser welding process, at **1200**, in the third stage, tail ends **1032** of respective protruding posts **1012** can be deformed to secure the first metal layer **1008** to the formed central thermal die pad **1018** to provide a structure, as shown in FIG. **12**. In some examples, the structure, as shown in FIG. **12** can be referred to as a leadframe assembly and thus can correspond to the leadframe assembly **300**, as shown in FIG. **3**.

(53) As shown in FIG. **13**, at **1300**, in a fourth stage, a bonding material **1034** can be deposited on the second side **1026** of the formed central thermal die pad **1018**. For example, the bonding material **1034** may be a polymeric type of material for attaching a chip **1036**. As shown in FIG. **14**, at **1400**, in a fifth stage, the chip **1036** can be attached to the bonding material **1034** to secure the chip **1036** to the second side **1026** of the formed central thermal die pad **1018**. As shown in FIG. **15**, at **1500**, in a sixth stage, at least one wire **1038** can be used to electrically couple a respective conductive terminal of the conductive terminals **1020** to a respective chip bonding pad **1040** of the chip **1036**. For clarity and brevity purposes, the example of FIG. **15** shows a single coupling between the respective conductive terminal of the conductive terminals **1020** and the respective chip bonding pad of the chip **1036**. However, it is be understood that any number of wires (e.g., copper wires) may be used to couple a corresponding chip bonding pad of the chip **1036** and a corresponding conductive terminal of the conductive terminals **1020**.

(54) As shown in FIG. **16**, at **1600**, in a seventh stage, a structure, as shown in FIG. **14**, can be encapsulated by a molding compound **1042** to provide a packaged semiconductor device **1044**, as shown in FIG. **16**. The packaged semiconductor device **1044** can correspond to the semiconductor device **100**, as shown in FIG. **1**. As an example, the molding compound **1042** can be an epoxy

based molding compounding.

(55) FIGS. **17-24** show stages of a method for fabricating a semiconductor device, such as the semiconductor device **200**, as shown in FIG. **2**. Thus, reference can be made to the example of FIGS. **2** and **7** in the example of FIGS. **17-24**. As shown in FIG. **17**, at **1700**, in a first stage, openings **1702** can be formed on a first side **1704** opposite a second side **1706** of a first metal layer **1708**. In some examples, the openings **1702** are trench openings. In some examples, the first metal layer **1708** can be referred to as a heat slug (e.g., the heat slug **210**, as shown in FIG. **2**). The first metal layer **1708** may be a conductive metal, such as Copper. In some examples, the openings **1702** are formed by performing an etch on the first side **1704** of the first metal layer **1708**. By way of example, a mask can be deposited on the first side **1704** of the first metal layer **1708** and etched to expose the openings **1702**. The mask thereafter can be stripped away to result in a structure as shown in FIG. **17**.

(56) As shown in FIG. **18**, at **1800**, in a second stage, a second metal layer **1710** can be patterned to provide a leadframe **1712** that includes a formed central thermal die pad **1714** and conductive terminals **1716**. The second metal layer **1710** may be a conductive metal, such as Copper or Copper alloy. In some examples, the first metal layer **1708** can have a greater thickness than the second metal layer **1710**. In other examples, the first and second metal layers **1708** and **1710** can have a similar thickness. For example, the leadframe **1712** can be produced, at **1800**, in the second stage, by a chemical etching technique or a stamping technique. As shown in FIG. **19**, at **1900**, in a third stage, a bonding material **1718** can be deposited on a first side **1720** opposite a second side **1722** of the formed central thermal die pad **1714**. For example, the bonding material **1718** may be a polymeric type of material for attaching the first metal layer **1708**. As shown in FIG. **20**, at **2000**, in a fourth stage, the first metal layer **1708** can be attached to the bonding material **1718** to secure the first metal layer **1708** to the first side **1720** of the formed central thermal die pad **1714**. The openings **1702** of the first metal layer **1708** can be used for alignment of the first metal layer **1708** with respect to the formed central thermal die pad **1714**, such that the openings **1702** are located near a periphery portion **1724** the formed central thermal die pad **1714**.

(57) As shown in FIG. **21**, at **2100**, in a fifth stage, a bonding material **1726** can be deposited on the second side **1722** of the formed central thermal die pad **1714**. For example, the bonding material **1726** may be a polymeric type of material for attaching a chip **1728**. As shown in FIG. **22**, at **2200**, in a sixth stage, the chip **1728** can be attached to the bonding material **1726** to secure the chip **1728** to the second side **1722** of the formed central thermal die pad **1714**. As shown in FIG. **23**, at **2300**, in a seventh stage, at least one wire **1730** can be used to electrically couple a respective conductive terminal of the conductive terminals **1716** to a respective chip bonding pad **1732** of the chip **1728**. For clarity and brevity purposes, the example of FIG. **23** shows a single coupling between the respective conductive terminal of the conductive terminals **1716** and the respective chip bonding pad of the chip **1728**. However, it is be understood that any number of wires (e.g., copper wires) may be used to couple a corresponding chip bonding pad of the chip **1728** and a corresponding conductive terminal of the conductive terminals **1716**.

(58) As shown in FIG. **24**, at **2400**, in an eighth stage, a structure, as shown in FIG. **24**, can be encapsulated by a molding compound **1734** to provide a packaged semiconductor device **1736**, as shown in FIG. **24**. The packaged semiconductor device **1736** can correspond to the semiconductor device **200**, as shown in FIG. **2**. As an example, the molding compound **1734** can be an epoxy based molding compounding.

(59) What have been described above are examples. It is, of course, not possible to describe every conceivable combination of components or methodologies, but one of ordinary skill in the art will recognize that many further combinations and permutations are possible. Accordingly, the disclosure is intended to embrace all such alterations, modifications, and variations that fall within the scope of this application, including the appended claims.

(60) Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

Claims

1. A method, comprising: attaching a front side of a metal layer to a die pad of a leadframe comprising conductive terminals, so a periphery portion of the metal layer extends beyond a periphery pad surface of the pad, and a portion of a half-etched cavity on the front side of the metal layer is located near the periphery pad surface of the pad; attaching a semiconductor device to the pad; and encapsulating the semiconductor device, the front side of the metal layer, a portion of a back side of the metal layer, and a portion of the conductive terminals to form a packaged semiconductor device.
2. The method of claim 1, wherein the half-etched cavity on the front side of the metal layer comprises protruding posts after cavity etching on the front side of the metal layer, wherein the protruding posts are located within the half-etched cavity.
3. The method of claim 2, the method further comprising patterning a second metal layer according to a pattern to provide the leadframe, wherein the patterning comprises etching through holes in the die pad.
4. The method of claim 3, wherein attaching the front side of the metal layer to the die pad comprises: inserting each of the protruding posts through a respective through hole of the through holes, such that a distal end of each protruding post extends through the respective through hole; and pressing the distal end of each protruding post to attach the metal layer to the die pad.
5. The method of claim 3, wherein attaching the front side of the metal layer to the die pad comprises applying a laser welding process to secure the first metal layer to the die pad after positioning each of the protruding posts through a respective through hole of the through holes.
6. The method of claim 5, wherein after the first metal layer is secured to the die pad, the method further comprises electrically coupling the portion of the conductive terminals to respective conductive terminals of the semiconductor device.
7. The method of claim 1, wherein the packaged semiconductor device is a leadless packaged semiconductor device.
8. A method, comprising: etching grooves on a front side of a metal layer opposite of a back side of the metal layer; attaching the front side of the metal layer to a die pad of a leadframe comprising conductive terminals, so a periphery portion of the metal layer extends beyond a periphery pad surface of the die pad, and the etched grooves are located near the periphery of the die pad; attaching a semiconductor device to the formed central thermal die pad; and encapsulating the semiconductor device, the front side of the metal layer, a portion of the back side of the metal layer, and a portion of the conductive terminals to form a packaged semiconductor device.
9. The method of claim 8, wherein the grooves are half-etched grooves, and the attaching the metal layer to the die pad comprises: depositing a layer of a bonding material on a surface of the die pad; and bonding using the bonding material the front side of the metal layer to a surface of the die pad.
10. The method of claim 9, wherein the surface of the die pad is a first surface, and the die pad has a second surface, and the bonding material is a first bonding material, the method further comprising: depositing a layer of a second bonding material on the second surface of the die pad; and bonding using the second bonding material the semiconductor device to the second surface of the die pad.
11. The method of claim 10, further comprising electrically coupling the portion of the conductive terminals to the semiconductor device after bonding the semiconductor device to the second surface of the die pad.
12. The method of claim 11, wherein the metal layer is a first metal layer, the method further comprising patterning a second metal layer according to a pattern to provide the leadframe.

13. The method of claim 8, wherein the packaged semiconductor device is a leadless packaged semiconductor device.
14. A semiconductor device comprising: a leadframe comprising a die pad having a pad surface; a chip attached to the die pad; and a metal layer directly connected to the die pad, the metal layer having first and second opposite sides, the metal layer comprising grooves extending into the metal layer from the first opposite side of the metal layer toward, but not touching, the second opposite side of the metal layer, wherein a periphery portion of the metal layer extends beyond a periphery pad surface of the die pad, and the grooves are located near the periphery of the die pad.
15. The semiconductor device of claim 14, wherein the leadframe comprises conductive leads, the semiconductor device further comprising a package layer configured to encapsulate the chip, the first side of the metal layer, and a portion of the conductive leads.
16. The semiconductor device of claim 15, wherein at least some of the portion of the conductive leads are separated from the die pad via a gap and the grooves are partially located within the gap.
17. The semiconductor device of claim 16, wherein the leadframe comprises conductive leads, the semiconductor device further comprising wires coupling respective conductive leads to the chip.
18. The semiconductor package of claim 17, wherein the semiconductor device is one of a Quad Flat No-Lead (QFN) semiconductor device and a Small Outline No-Lead (SON) semiconductor device.
19. A semiconductor device comprising: a leadframe comprising a die pad, wherein the die pad comprises through holes extending through the die pad; a chip attached to the die pad; and a metal layer comprising protruding posts on a front side of the metal layer opposite of a back side of the metal layer, in which the protruding posts extend into respective through holes of the die pad, and a periphery portion of the metal layer extends beyond a periphery pad surface of the die pad.
20. The semiconductor device of claim 19, wherein the leadframe comprises conductive leads, the semiconductor device further comprising a molding compound encapsulating the chip and the front side of the metal layer.
21. A semiconductor device comprising: a leadframe comprising a die pad and a conductive lead, wherein the die pad comprises through holes extending through the die pad; a chip attached to the die pad; a metal layer comprising protruding posts on a front side of the metal layer opposite of a back side of the metal layer, in which the protruding posts extend into respective through holes of the die pad, and a periphery portion of the metal layer extends beyond a periphery pad surface of the die pad, and wherein the metal layer further comprises a half-etched cavity on the front side of the metal layer and a portion of the half-etched cavity is located near a periphery of the die pad; and a molding compound encapsulating the chip and the front side of the metal layer.
22. The semiconductor device of claim 21, wherein the semiconductor device is one of a Quad Flat No-Lead (QFN) semiconductor device and a Small Outline No-Lead (SON) semiconductor device.
23. The semiconductor device of claim 22, wherein the metal layer is secured to the die pad via one of a riveting process or a welding process opposite of a side to which the chip is attached.
24. The semiconductor package of claim 23, wherein the patterned leadframe comprises conductive leads, the semiconductor device further comprising wires coupling the portion of the leads to the semiconductor device.
25. The semiconductor device of claim 14, wherein rivets attach the metal layer to the die pad.
26. The semiconductor device of claim 14, wherein the metal layer is welded to the die pad.
27. The semiconductor device of claim 14, wherein the metal layer is thicker than the die pad.
28. The semiconductor device of claim 14, wherein the metal layer is the same thickness as the die pad.
29. The semiconductor device of claim 14, further including a molding compound encapsulating the chip and the front side of the metal layer.
30. The semiconductor device of claim 19, wherein distal ends of the protruding posts are deformed to secure the metal layer to the die pad.

31. The semiconductor device of claim 19, wherein the metal layer is welded to the die pad.
 32. The semiconductor device of claim 19, wherein the metal layer is thicker than the die pad.
 33. The semiconductor device of claim 19, wherein the metal layer is the same thickness as the die pad.
 34. The semiconductor device of claim 19, further including a molding compound encapsulating the chip and the front side of the metal layer.
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